



DIPARTIMENTO
DI INGEGNERIA
DELL'INFORMAZIONE

1222 · 2022
800
ANNI



UNIVERSITÀ
DEGLI STUDI
DI PADOVA

Project Proposals

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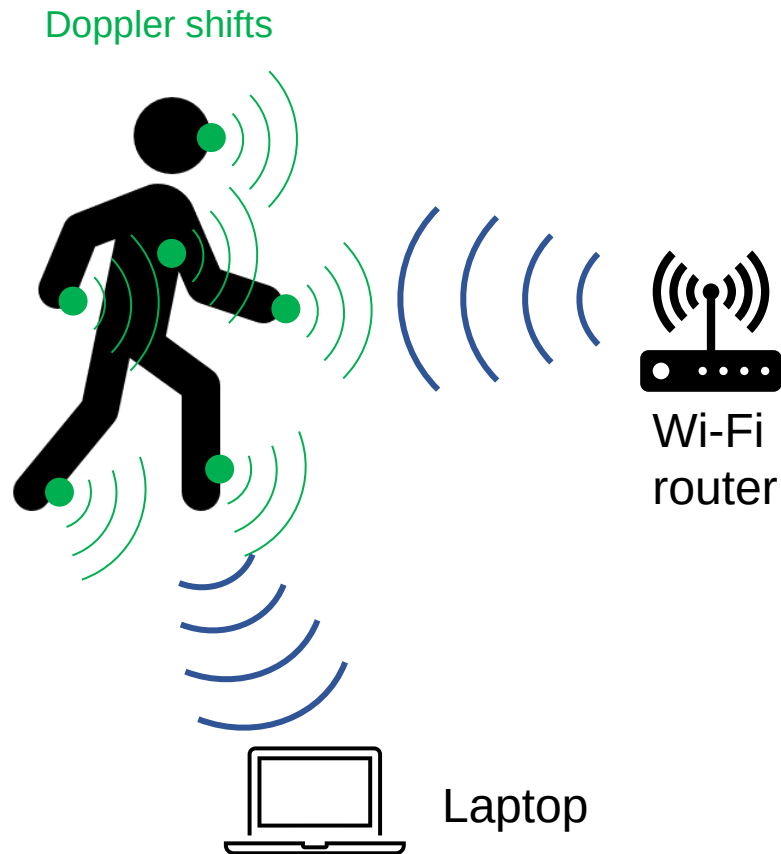
Human Activity Recognition with Wi-Fi

Reference papers

- [1] F. Meneghello, D. Garlisi, N. D. Fabbro, I. Tinnirello and M. Rossi, "SHARP: Environment and Person Independent Activity Recognition With Commodity IEEE 802.11 Access Points," in *IEEE Transactions on Mobile Computing*, vol. 22, no. 10, pp. 6160-6175, 1 Oct. 2023.
- [2] F. Meneghello, N. D. Fabbro, D. Garlisi, I. Tinnirello and M. Rossi, "A CSI Dataset for Wireless Human Sensing on 80 MHz Wi-Fi Channels," in *IEEE Communications Magazine*, vol. 61, no. 9, pp. 146-152, September 2023.
- [3] GitHub: <https://github.com/francescamen/SHARP>

Wireless sensing

- Contactless sensing of human activities is extremely important in many applications
- Healthcare, security, monitoring
- Wi-Fi routers devices can be repurposed to act as radars to monitor human movement with high accuracy
- No video is recorded

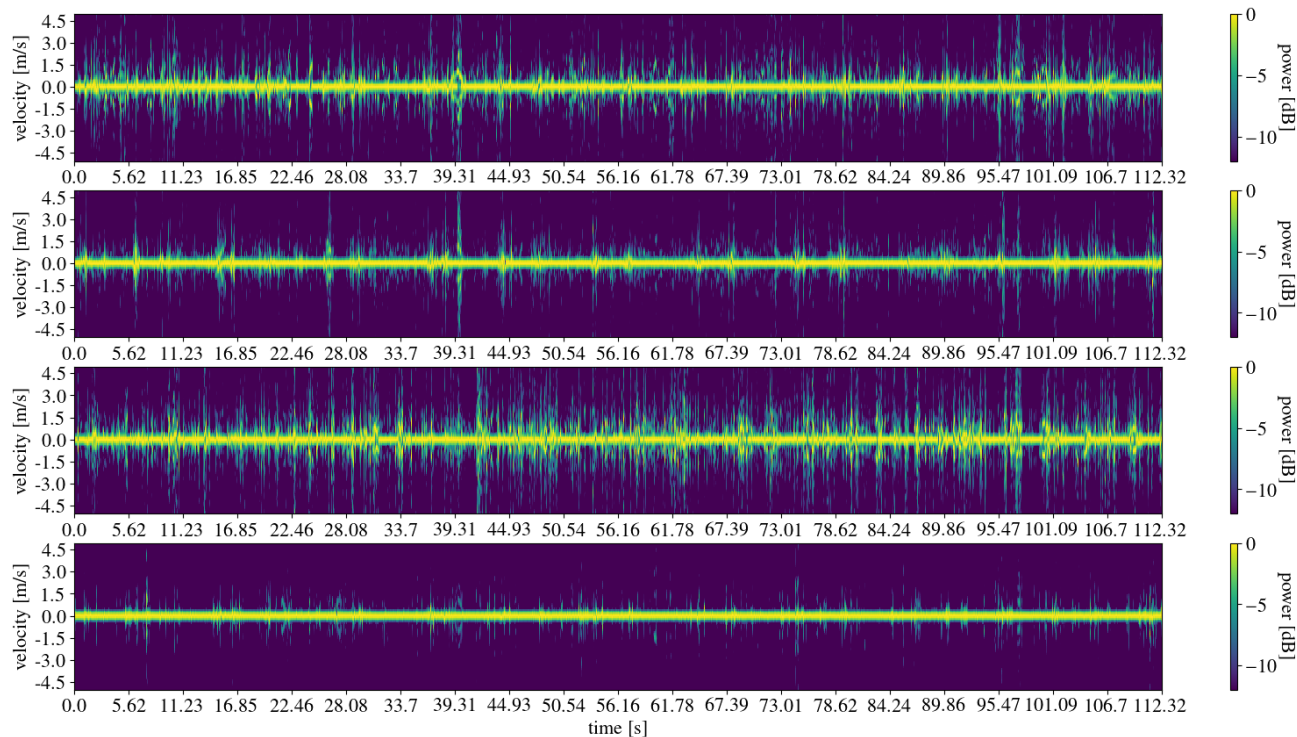


Doppler spectrograms from Wi-Fi packets

Obtained via Short-Time Fourier Transform of the channel estimated by the receiver

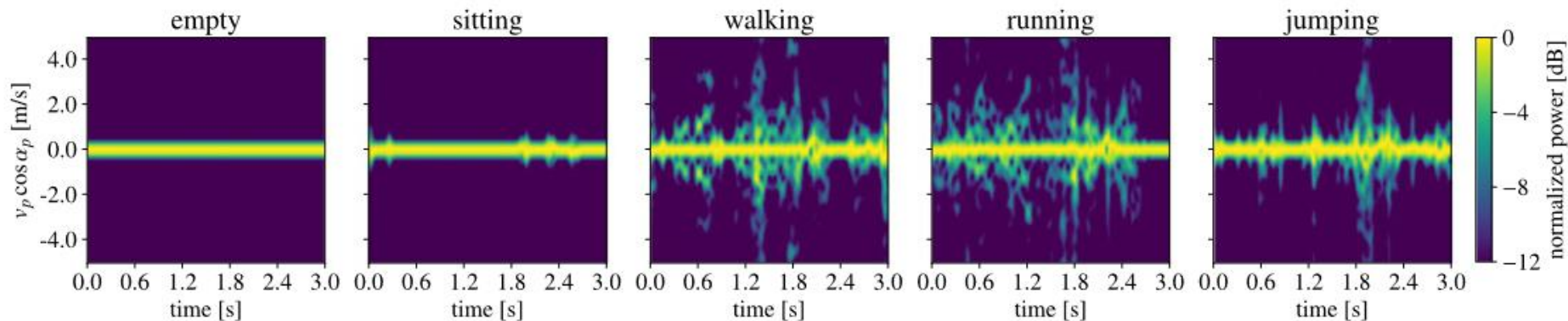
They change significantly for each activity performed by the subject

They can be treated as “images” and processed via a CNN



Doppler spectrogram vs activity

- They change significantly for each activity performed by the subject
- They can be treated as “images” and processed via a CNN



The dataset

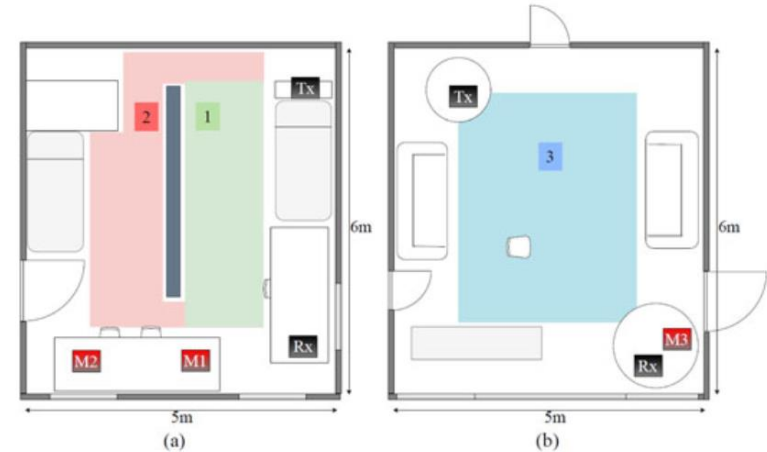
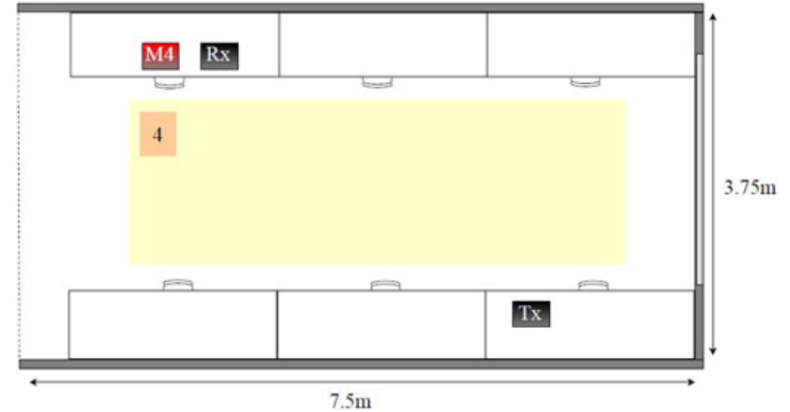
- Link: <https://ieee-dataport.org/documents/csi-dataset-wireless-human-sensing-80-mhz-wi-fi-channels>
 - Only accessible for IEEE members
 - But contains key information on the data
- **Pre-extracted Doppler traces at [this link](#)**
 - 7 different activities
 - **10** different people
 - 7 different environments

Diversity is key

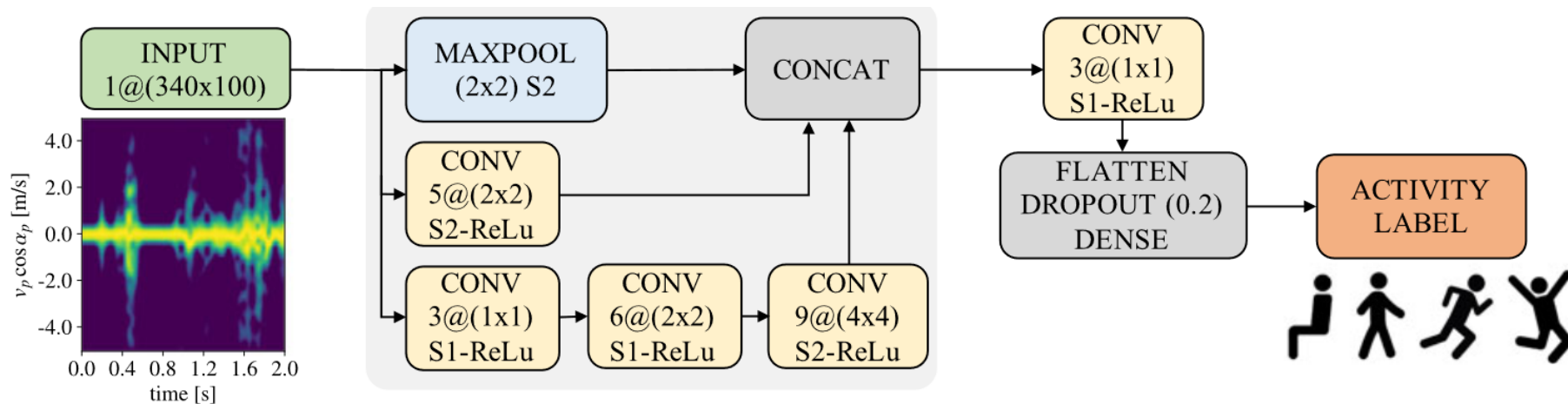
- Environments
- People

TABLE 1
Measurement Conditions

set	monitor position, M_j	person, P_i	direct path	type
S_1	M1	P1	yes	train-test
S_2	M1	P1	yes	test
S_3	M1	P2	yes	test
S_4	M2	P1	no	test
S_5	M2	P2	no	test
S_6	M3	P1	yes	test
S_7	M4	P3	yes	test



Baseline model



Challenges and possible objectives

1. Different subjects act differently

- Generalizing to different subjects is challenging
- Contrastive/self supervised approach?

2. Different environments modify the signal differently

- Performance could drop when the room changes
- Even across multiple days

3. Person identification

- Exploit point 1. to recognize the identity of the subject

